Projective Bottlenecks in Passive Spectral Routing: Exact Planar Memory, Quantitative Approximation Barriers, and Cross-Ratio Strata

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Abstract

We determine the exact passive state cost of every finite routing table whose target lines lie in one complex two-plane. At distinct boundary frequencies ζ_i, a square rational-inner matrix must route one input line to prescribed target lines Y_i. Let δ be the least algebraic degree of a rational map $\mathbb{P}^1 \to \mathbb{P}^1$ interpolating the ordered projective data (ζ_i, Y_i). We prove the exact reduction

$$d_{\min} = \delta.$$

The upper bound is constructive: scalar Fejér–Riesz factorization lifts any coprime rational pair to an explicit 2×2 lossless network of exactly the same McMillan degree. The lower bound uses a common state-space denominator: more boundary zeros than states force the routed column to remain identically inside the target plane, where it becomes a projective rational interpolant.

The same denominator mechanism yields a quantitative result for approximate routing. If the normalized interpolation matrix M_d has smallest singular value $\sigma > 0$, then every lossless router of degree at most d has worst-node complex-projective chordal error at least

$$\frac{\sigma}{\sqrt{L(d+1) + \sigma^2}}.$$

Under a data-matrix perturbation of norm η, the bound survives with σ replaced by $(\sigma - \eta)_+$. Thus the algebraic obstruction has a finite-error consequence, not merely an exact-interpolation discontinuity.

This reduction reveals two competing mechanisms. For two target lines with occupancies n_0, n_∞, the exact cost is $\max(n_0, n_\infty)$, a fiber multiplicity law. By contrast, for L distinct generic coplanar targets,

$$d_{\min} = \left\lceil \frac{L-1}{2} \right\rceil,$$

whereas both the span bound and the optimized constant-detector hierarchy of the preceding framework equal one. Their multiplicative underestimate is therefore unbounded. For four nodes we give the complete phase diagram: degree zero for one constant target, degree one exactly for four distinct targets with matching ordered cross-ratio, degree three for a $3+1$ collision, and degree two in every remaining case. For the strict four-node fixture, an exact Sylvester certificate forces more than 21% worst-node error on every one-state router. Exact Gaussian-rational fixtures verify all strata, the quantitative obstruction, and the generic law through $L = 16$. Exact producer, replay validator, frozen evidence, and immutable release metadata are linked in the paper.

Keywords: rational inner matrix; projective interpolation; McMillan degree; passive memory; approximate routing; chordal error; cross-ratio; Fejér–Riesz factorization; Wigner–Smith delay.

1 A singular stratum with hidden memory

Let S be an $N \times N$ rational matrix, analytic in $\mathbb{D}$, unitary almost everywhere on $\mathbb{T}$, and regular at distinct nodes $\zeta_1, \ldots, \zeta_L \in \mathbb{T}$. Its McMillan degree is the state dimension of a minimal conservative realization. Fix an input line $X \subset \mathbb{C}^N$ and target lines $Y_i \subset \mathbb{C}^N$. The routing task is

$$S(\zeta_i)X = Y_i, \qquad 1 \le i \le L. \tag{1}$$

Endpoint phases and bases are free. Write $d_{\min}$ for the least McMillan degree among all such S.

The generic direct-sum law counts states by target occupancy, and its constant-detector extension gives useful lower bounds on singular target arrangements [1]. It deliberately leaves open a complete equality theory outside direct-sum position. The first unresolved stratum in that preceding framework is also the smallest geometrically nontrivial one: all Y_i lie in a fixed two-plane H. Here the span statistic is frozen, yet projective compatibility can demand arbitrarily many states.

Classical scalar rational interpolation and its state-variable realization describe minimal rational functions through finite data [2, 3, 4, 5]. Minimal boundary interpolation by scalar unimodular functions and finite Blaschke products is also well developed [6, 7, 8]. Subspace Nevanlinna interpolation and boundary matrix all-pass construction provide the matrix context [9, 10, 11]. Our contribution is not a new scalar interpolation algorithm. It is the exact identification of that scalar projective degree with the physical state count of a square passive lossless router, together with a quantitative approximation barrier for low-state routers, the resulting generic law, unbounded detector gap, and complete four-line stratification.

2 Projective degree equals passive memory

Choose a basis of H and identify its lines with $\mathbb{P}(H) \simeq \mathbb{P}^1$. A rational map $R : \mathbb{P}^1 \to \mathbb{P}^1$ is represented by a coprime polynomial pair $[p : q]$, and $\deg R = \max(\deg p, \deg q)$. Poles of the affine ratio p/q are ordinary points of the projective map, not physical singularities.

Definition 2.1 (Projective interpolation degree)**.** For the ordered data (ζ_i, Y_i), let

$$\delta = \min\{\deg R : R : \mathbb{P}^1 \to \mathbb{P}(H),\ R(\zeta_i) = Y_i \text{ for every } i\}.$$

Common-denominator lemma. Let S be a rational-inner matrix of McMillan degree at most d, and let $F = Sx$ be any routed column. There are a polynomial χ and vector polynomial n such that

$$F(z) = \frac{n(z)}{\chi(z)}, \qquad \deg \chi \le d, \qquad \deg n_j \le d.$$

The denominator can be chosen nonzero at every regular interpolation node. Indeed, take a stable minimal realization $S(z) = D + zC(I - zA)^{-1}B$ of dimension at most d. With $\chi(z) = \det(I - zA)$, the adjugate formula gives the stated degree bounds for every entry of Sx. Stability makes χ zero-free on $\overline{\mathbb{D}}$; more generally, after cancelling any removable common factor, it is nonzero at each node where S is regular.

Theorem 2.2 (Exact planar reduction)**.** *If all target lines in* (1) *lie in one two-plane H, then*

$$\boxed{d_{\min} = \delta.} \tag{2}$$

The equality is independent of the ambient dimension and is constructive.

Proof. We first note that $\delta \le L - 1$. Apply a projective change of target chart whose point at infinity avoids the finite target set, interpolate the resulting affine values by a polynomial of degree at most $L - 1$, and undo the chart.

For the upper bound, let $[p : q]$ be a coprime representative of degree δ. For a polynomial r of degree at most δ, set

$$r^\sharp(z) = z^\delta \overline{r(1/\bar{z})}.$$

Because p and q are coprime, they have no common zero; in particular $|p|^2 + |q|^2 > 0$ on $\mathbb{T}$. Scalar Fejér–Riesz factorization [12] supplies an outer polynomial h, zero-free in $\mathbb{D}$, such that on $\mathbb{T}$

$$|h|^2 = |p|^2 + |q|^2.$$

Equivalently, $h^\sharp h = p^\sharp p + q^\sharp q$. Therefore

$$S_2(z) = \frac{1}{h(z)} \begin{pmatrix} p(z) & -q^\sharp(z) \\ q(z) & p^\sharp(z) \end{pmatrix} \tag{3}$$

is analytic in $\mathbb{D}$ and unitary on $\mathbb{T}$. Its first column represents $[p : q]$, and

$$\det S_2 = h^\sharp / h.$$

The strict positivity above also gives $h(\zeta_i) \neq 0$ at every routing node. With reversal taken at the displayed order δ, the scalar Blaschke product $h^\sharp/h$ has degree exactly δ; if $\deg h < \delta$, the missing reversed zeros occur at zero. For a rational-inner matrix the determinant winding equals its McMillan degree [13], so $\deg_{\mathrm{McM}} S_2 = \delta$. Constant unitaries align the input line and the basis of H, while a block identity embeds S_2 into dimension N without adding states. Hence $d_{\min} \leq \delta$.

Conversely, take a minimum-degree interpolant S with degree d. The upper construction already gives $d \leq L - 1 < L$. A stable conservative realization of S and the common-denominator lemma above give every entry of the routed column $F = Sx$ a common denominator of degree at most d, nonzero at the boundary nodes; the corresponding numerators have degree at most d. For every linear functional ℓ annihilating H, the numerator of ℓF vanishes at all $L > d$ distinct nodes. It is therefore zero identically. Thus $F(z) \in H$ as a rational identity. In a basis of H, cancel the common polynomial factor of the two coordinate numerators of F. The resulting coprime pair has degree at most d and interpolates every Y_i, so $\delta \leq d$. This proves (2). □

Remark 2.3 (Affine poles are harmless). The datum $R(z) = 1/z$ has an affine pole at zero, but the projective pair $[1 : z]$ is regular there. Formula (3) lifts this pair to an analytic inner column. Accordingly, pole tests on a chosen affine ratio cannot replace projective coprimality and Fejér–Riesz lifting.

3 Exact certificates and quantitative approximation barriers

Write $Y_i = [a_i : b_i]$ in the chosen basis of H. For $d \geq 0$, define the $L \times 2(d+1)$ matrix

$$M_d(i,:) = \big(b_i, b_i\zeta_i, \ldots, b_i\zeta_i^d, -a_i, -a_i\zeta_i, \ldots, -a_i\zeta_i^d\big). \tag{4}$$

A coefficient vector in $\ker M_d$ is exactly a polynomial pair (p, q) of degree at most d satisfying $[p(\zeta_i) : q(\zeta_i)] = [a_i : b_i]$, provided it has no base point at a node.

Proposition 3.1 (Fail-closed degree certificate). *The number δ is the least d for which* $\ker M_d$ *contains a pair (p, q) generating the unit ideal, equivalently* $\gcd(p, q) = 1$. *For a nonconstant pair this can be certified by* $\mathrm{Res}(p, q) \neq 0$*; constant pairs are checked directly. Such a pair is an exact upper certificate. Full column rank of M_{d-1} is an exact lower certificate.*

Proof. The unit-ideal condition is precisely projective coprimality. When both polynomials form a nonconstant pair, it is equivalent to nonvanishing of the resultant; the constant pairs $[c : 0]$ and $[0 : c]$, $c \neq 0$, are checked by the unit-ideal condition itself. The claim follows from Definition 2.1 and (4). Full column rank excludes every nonzero pair of degree at most $d - 1$. □

The formulation is deliberately algebraic. A kernel vector alone is not enough: a common factor can inflate its displayed degree or introduce a base point. A Bézout identity $up + vq = 1$ is an equivalent independently checkable coprimality certificate.

We now normalize (a_i, b_i) to unit norm and choose an orthonormal basis of H. For a unit vector u and a target line $Y = \mathbb{C}y$, $\|y\| = 1$, write

$$\mathrm{dist}_{\mathrm{ch}}(\mathbb{C}u, Y) = \sqrt{1 - |\langle u, y\rangle|^2}$$

for complex-projective chordal distance.

Theorem 3.2 (Quantitative approximate-routing obstruction)**.** *Suppose M_d has full column rank and put $\sigma = \sigma_{\min}(M_d) > 0$. Every square rational-inner router S of McMillan degree at most d satisfies*

$$\max_{1\le i\le L} \mathrm{dist}_{\mathrm{ch}}\big(S(\zeta_i)X, Y_i\big) \;\ge\; \boxed{\frac{\sigma}{\sqrt{L(d+1)+\sigma^2}}}. \tag{5}$$

If a perturbed normalized data matrix $\widehat{M}_d$ obeys $\|\widehat{M}_d - M_d\|_2 \le \eta < \sigma$, the right-hand side remains valid with σ replaced by $\sigma - \eta$. Equivalently, the degree-constrained minimax error $E_d = \inf_{\deg_{\mathrm{McM}} S\le d} \max_i \mathrm{dist}_{\mathrm{ch}}(S(\zeta_i)X, Y_i)$ obeys the same lower bound.

Proof. Choose a unit input vector x and write $F = Sx$. On $\mathbb{T}$, $\|F\| = 1$. By the common-denominator lemma above, the two coordinates of $P_H F$ have the form $p/\chi, q/\chi$, with numerator degrees at most d. Let c collect their $2(d+1)$ coefficients. If $c = 0$, then F is orthogonal to every Y_i and the left-hand side of (5) is one, so assume $c \ne 0$.

Put $v_i = (p(\zeta_i), q(\zeta_i))$ and $e_i = \mathrm{dist}_{\mathrm{ch}}(\mathbb{C}F(\zeta_i), Y_i)$. Since $v_i = \chi(\zeta_i)P_H F(\zeta_i)$ and $y_i \in H$, the two-dimensional wedge identity gives

$$|(M_d c)_i| = |\det(y_i, v_i)| \le |\chi(\zeta_i)|e_i, \qquad \|v_i\| \ge |\chi(\zeta_i)|\sqrt{1-e_i^2}.$$

Evaluation of two degree-d polynomials at a unit-modulus node has operator norm $\sqrt{d+1}$, hence $\|v_i\| \le \sqrt{d+1}\|c\|$. If $\varepsilon = \max_i e_i < 1$, these inequalities imply

$$\|M_d c\|_2 \le \sqrt{L(d+1)}\,\|c\|\,\frac{\varepsilon}{\sqrt{1-\varepsilon^2}}.$$

The reverse inequality $\|M_d c\|_2 \ge \sigma\|c\|$ and cancellation of $\|c\|$ give (5); the case $\varepsilon = 1$ is immediate. Finally, Weyl's inequality gives $\sigma_{\min}(\widehat{M}_d) \ge \sigma - \eta$, proving the perturbative claim. □

Corollary 3.3 (Exact-arithmetic coercivity certificate)**.** *Let $G_d = M_d^* M_d$, $m = 2(d+1)$, and*

$$\gamma_d = \det G_d \left(\frac{m-1}{\operatorname{tr} G_d}\right)^{m-1}.$$

If M_d has full column rank, then $\sigma^2 \ge \gamma_d > 0$, so the right-hand side of (5) is at least

$$\sqrt{\frac{\gamma_d}{L(d+1)+\gamma_d}}.$$

For Gaussian-rational nodes and targets, γ_d is an exact arithmetic certificate. Any stronger exact inequality $G_d \succeq \alpha I$ replaces γ_d by α.

Proof. If $0 < \lambda_1 \le \cdots \le \lambda_m$ are the eigenvalues of G_d, then $\det G_d = \lambda_1 \prod_{j>1} \lambda_j$. By arithmetic–geometric mean,

$$\prod_{j>1} \lambda_j \le \left(\frac{\sum_{j>1}\lambda_j}{m-1}\right)^{m-1} \le \left(\frac{\operatorname{tr} G_d}{m-1}\right)^{m-1}.$$

Thus $\sigma^2 = \lambda_1 \ge \gamma_d$. Apply Theorem 3.2. □

The normalization is essential: without unit homogeneous representatives and an orthonormal basis of H, σ changes under coordinate rescaling. The theorem now controls routing error itself, rather than only persistence of an exact rank obstruction.

4 Fiber multiplicity and binary routing

Projective interpolation makes collisions transparent. If a nonconstant rational map has degree d, no fiber contains more than d distinct points. This observation becomes a state lower bound after Theorem 2.2.

Proposition 4.1 (Fiber occupancy bound)**.** *Suppose at least two distinct target lines are used, and let* $n_{\max} = \max_Y \#\{i : Y_i = Y\}$. *Then*

$$d_{\min} \geq n_{\max}.$$

Proof. A constant projective map cannot hit two targets. For a nonconstant map $R = [p : q]$ of degree d, the equation $R(z) = Y$ is a nonzero polynomial equation of degree at most d. It has at most d distinct solutions. Apply this to a most occupied target and use Theorem 2.2. □

For exactly two targets, the fiber obstruction is the whole answer.

Theorem 4.2 (Exact binary collision law)**.** *If the table uses exactly two distinct target lines with occupancies n_0 and n_∞, then*

$$d_{\min} = \max(n_0, n_\infty). \tag{6}$$

Proof. Normalize the two targets to 0 and ∞. Let I_0, I_∞ be their node sets. The coprime pair

$$p(z) = \prod_{i \in I_0} (z - \zeta_i), \qquad q(z) = \prod_{i \in I_\infty} (z - \zeta_i)$$

has degree $\max(n_0, n_\infty)$ and realizes the table. The reverse inequality is Theorem 4.1. □

The binary law is the collision-dominated extreme: the constant-detector bound is sharp. The opposite extreme has all targets distinct. Then fiber multiplicity contributes only one, but global projective compatibility produces a growing cost.

Table 1: Exact planar mechanisms. The projective reduction converts every entry in the last column directly into passive state count.

Target pattern	Exact or generic degree	Governing obstruction
One target	0	constant projective map
Two targets, occupancies n_0, n_∞	$\max(n_0, n_\infty)$	largest fiber
L distinct, generic	$\lceil (L-1)/2 \rceil$	projective parameter count
Four distinct, exceptional	1 or 2	ordered cross-ratio
Four targets, pattern $3+1$	3	triple fiber

5 The generic planar law and an unbounded gap

Lemma 5.1 (Nonempty coprime threshold locus)**.** *Fix distinct nodes and put $d_0 = \lceil (L-1)/2 \rceil$. There is a nonempty Zariski-open set of target data for which M_{d_0} contains a coprime kernel pair. This open set meets the locus of pairwise distinct targets.*

Proof. Work first in the affine target chart $a_i/b_i = y_i$ and use the witness $y_i = \zeta_i^{d_0}$. At this point the columns of M_{d_0} are evaluations of monomials with exponents $0, \dots, 2d_0$, with exponent d_0 duplicated. Because $L \leq 2d_0 + 1$, a Vandermonde $L \times L$ minor D is nonzero. On the principal open set $U = \{D \neq 0\}$, this same pivot solves L coefficients by Gaussian elimination and expresses the remaining coefficients as free parameters with entries in the localization $\mathbb{C}[y_1, \dots, y_L]_D$. Hence $\ker M_{d_0}$ has an algebraic frame on U.

At the witness, $(z^{d_0}, 1)$ lies in the kernel. Choose the constant linear combination of the local frame that equals this vector there. Multiplying its two polynomial coordinates by a common power of D clears every parameter denominator. Their resultant is then a regular function of the target data; at the witness it is a nonzero scalar multiple of $\mathrm{Res}(z^{d_0}, 1) = 1$. Its nonvanishing and $D \neq 0$ therefore define a nonempty Zariski-open coprime locus. The pairwise target inequalities also define a nonempty Zariski-open subset of $(\mathbb{P}^1)^L$. Their intersection is nonempty because $(\mathbb{P}^1)^L$ is irreducible. □

Theorem 5.2 (Generic planar memory). *For fixed distinct nodes and L target lines in a two-plane, there is a Zariski-open dense set of ordered target data on which*

$$d_{\min} = \left\lceil \frac{L-1}{2} \right\rceil. \tag{7}$$

The open set can be intersected with the locus of pairwise distinct targets.

Proof. Put $d_0 = \lceil (L-1)/2 \rceil = \lfloor L/2 \rfloor$. For $d < d_0$, the matrix M_d has at most L columns and is generically full-column-rank. A concrete rank witness is obtained from affine data $a_i/b_i = \zeta_i^{d_0}$: an equation $p(\zeta_i) = \zeta_i^{d_0} q(\zeta_i)$ at all nodes would give L roots to a polynomial of degree at most $L-1$ and then force $p = q = 0$ at degree $d_0 - 1$. Thus the nonvanishing of a maximal minor supplies the required nonempty open set.

At $d = d_0$, M_{d_0} has more columns than rows, so it always has nonzero kernel. Lemma 5.1 supplies a nonempty open locus on which that kernel contains a coprime pair, and the lower-rank open condition above can be intersected with it. Apply Theorem 2.2 and Proposition 3.1. The scalar threshold itself is classical; the new conclusion here is its exact conversion to passive state count. □

For distinct lines in a two-plane, the span bound is $\dim H - 1 = 1$. Likewise, the optimized constant-detector hierarchy from [1] equals one: the kernel of a nonzero functional on H contains at most one of the distinct target lines.

Corollary 5.3 (Unbounded detector gap). *On the generic pairwise-distinct planar locus,*

$$\frac{d_{\min}}{\max\{\textit{span bound, constant-detector bound}\}} = \left\lceil \frac{L-1}{2} \right\rceil \longrightarrow \infty.$$

Thus the optimized constant-detector hierarchy of [1] cannot approximate the exact passive memory within a universal multiplicative factor on this single fixed-rank stratum.

6 The complete four-line phase diagram

For four ordered distinct nodes, write

$$[z_1, z_2; z_3, z_4] = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}$$

with the usual projective conventions at infinity. The order matters because each frequency is paired with a specified target. The projective classification below is classical/elementary; our claim is its exact consequence for square passive rational-inner routers.

Theorem 6.1 (Four-line classification). *For four target lines in one two-plane, the exact passive memory is*

$$d_{\min} = \begin{cases} 0, & \textit{all four targets coincide}, \\ 1, & \textit{all four are distinct and their ordered cross-ratio equals that of the nodes}, \\ 3, & \textit{the target multiplicities are } 3+1, \\ 2, & \textit{in every remaining case.} \end{cases} \tag{8}$$

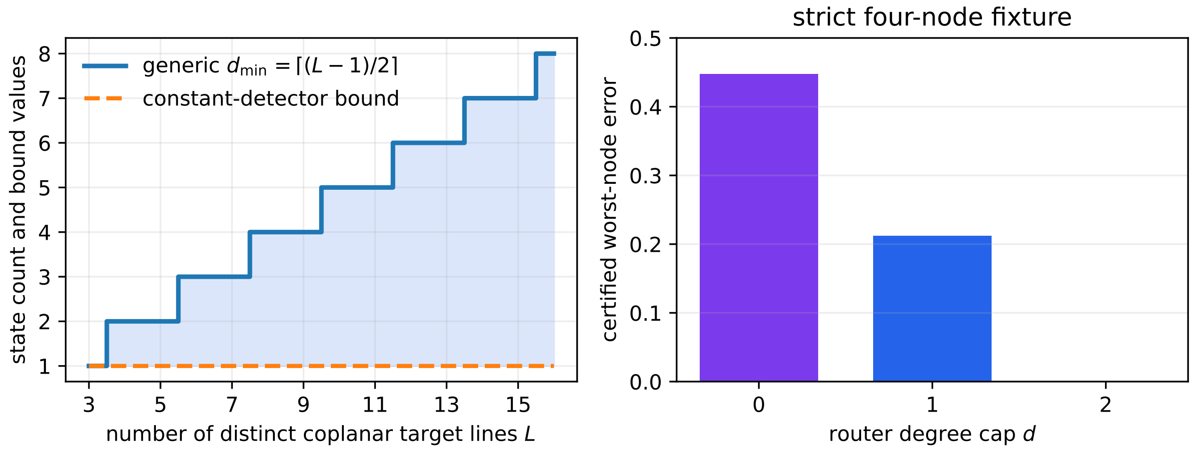

Figure 1: Left: generic exact memory versus the constant-detector lower bound; the multiplicative ratio is unbounded. Right: exact certified worst-node chordal-error barriers for the strict four-node fixture. Degree two attains zero error, whereas degrees zero and one remain quantitatively separated.

Proof. A nonconstant degree-one map is a projective bijection, so it sends the four nodes to four distinct targets and preserves their ordered cross-ratio. These conditions are also sufficient by the uniqueness of a projective map through three ordered points.

If three nodes share a target and the fourth does not, a nonconstant map of degree at most two cannot work because every fiber has at most two distinct points. Polynomial interpolation gives the universal upper bound three.

It remains to show degree at most two in all other nonconstant cases. For four distinct targets with mismatched cross-ratio, normalize domain and target triples to $(0, \infty, 1)$, leaving $s \mapsto t$ with $s \neq t$. Then

$$R(z) = z(az + b), \qquad a = \frac{t-s}{s(s-1)}, \quad b = 1 - a$$

has degree two and realizes the data. For multiplicities $2 + 2$, after choosing a finite chart use

$$R(z) = c\frac{(z - z_1)(z - z_2)}{(z - z_3)(z - z_4)}.$$

For $2 + 1 + 1$, normalize the repeated nodes to $0, \infty$, the other nodes to $1, s$, and the targets to $\infty, \infty, 0, 1$. Choose $a \notin \{0, 1, s\}$ and set

$$q(z) = z, \qquad p(z) = A(z-1)(z-a), \qquad A = \frac{s}{(s-1)(s-a)}.$$

This is a coprime degree-two witness. None of these cases is constant or degree one, completing the lower bounds and the classification via Theorem 2.2. □

Example 6.2 (Strict exact and approximate detector failure)**.** Take

$$(\zeta_1, \zeta_2, \zeta_3, \zeta_4) = (1, i, -1, -i), \qquad (Y_1, Y_2, Y_3, Y_4) = (0, 1, \infty, 2).$$

The ordered cross-ratios are 2 and 1/2, respectively. Hence $d_{\min} = 2$, while both constant lower bounds equal one. An exact witness is

$$p(z) = -\frac{(z-1)(3z-i)}{2}, \qquad q(z) = z + 1, \qquad \mathrm{Res}(p, q) = -3 - i \neq 0.$$

It takes the four requested projective values exactly.

There is also a finite approximation gap below degree two. Normalize the four target representatives to unit norm. For $d = 1$, exact arithmetic gives

$$G_1 = M_1^* M_1 = \begin{pmatrix} 17/10 & 1+3i/10 & -9/10 & -i/10 \\ 1-3i/10 & 17/10 & i/10 & -9/10 \\ -9/10 & -i/10 & 23/10 & -1-3i/10 \\ i/10 & -9/10 & -1+3i/10 & 23/10 \end{pmatrix}.$$

The leading principal minors of $G_1 - \frac{3}{8}I$ are

$$\frac{53}{40}, \quad \frac{213}{320}, \quad \frac{637}{2560}, \quad \frac{213}{20480},$$

all positive. Sylvester's criterion yields $G_1 \succ \frac{3}{8}I$, and Theorem 3.2 therefore forces

$$\max_i \operatorname{dist}_{\mathrm{ch}}(S(\zeta_i)X, Y_i) > \sqrt{\frac{3}{67}} \approx 0.2116$$

for every router with at most one state. Likewise, $G_0 - I \succ 0$ gives the degree-zero barrier $1/\sqrt{5} \approx 0.4472$. The degree-two witness above attains zero, producing a certified finite-error staircase rather than only an exact degree jump.

7 Delay price and reproducibility

For a differentiable lossless boundary response, define the Wigner–Smith matrix

$$Q(\theta) = -iS(e^{i\theta})^* \frac{\mathrm{d}}{\mathrm{d}\theta} S(e^{i\theta}).$$

Jacobi's formula gives $\operatorname{tr}(S^{-1}\partial_\theta S) = \partial_\theta \log \det S$. Together with Potapov's factorization and winding count, this gives

$$\int_0^{2\pi} \operatorname{tr} Q(\theta)\, \mathrm{d}\theta = 2\pi \deg_{\mathrm{McM}} S$$

for finite rational-inner S [14, 15]. Combining it with Theorem 2.2 yields an exact action law.

Corollary 7.1 (Optimal integrated delay). *For every planar line-routing table,*

$$\inf_S \int_0^{2\pi} \operatorname{tr} Q(\theta)\, \mathrm{d}\theta = 2\pi\delta.$$

The explicit completion (3) *attains the infimum.*

The computational artifact uses exact arithmetic over $\mathbb{Q}(i)$. It checks the strict witness in Example 6.2; exact representatives of the constant, cross-ratio-compatible, $2+2$, $2+1+1$, and $3+1$ strata; the normalized Gram matrices and Sylvester coercivity certificates behind the degree-zero and degree-one approximation barriers; full-column-rank exclusions below the generic threshold; coprime threshold witnesses for every $3 \le L \le 16$; and every nontrivial binary occupancy split through $L = 16$ (119 exact cases). The frozen validator recomputes the content hash and checks declared coverage, phase degrees, ranks, and bounds. In CI the producer is replayed from scratch and its byte-identical JSON is compared with the frozen record; this replay is the algebraic check of cross-ratios, witness values, gcds, and resultants. These calculations audit examples and certificates; they are not substitutes for the proofs.

Limitation 7.2 (Exact scope). The theorem assumes distinct frequency nodes, one input line, target lines in one common two-plane, free endpoint phases, and square finite rational-inner competitors. Repeated nodes require confluent interpolation. Fixed frames, higher-rank subspaces, lossy or active architectures, and a complete equality theory for nonplanar singular arrangements are not claimed. The quantitative obstruction is a computable lower bound, not a claim of sharpness for the full degree–error tradeoff; determining E_d exactly beyond the exhibited strata remains open.

Limitation 7.3 (Priority scope). Scalar minimal rational interpolation, the generic half-degree threshold, cross-ratio and fiber classifications, subspace Nevanlinna interpolation, Fejér–Riesz/paraunitary completion, and matrix all-pass construction predate this work. We claim the passive-memory equality (2), obtained by combining a standard exact-degree lossless lift with the escape-from-the-plane lower bound, the quantitative low-degree obstruction (5), and their physical/detector consequences; we do not claim a new general rational interpolation algorithm or priority for the scalar strata.

Immutable reproducibility record. The tagged release containing the paper, exact producer, frozen validator, frozen JSON certificate, and figure is:

GitHub release: `v3.1-planar-projective-memory`

8 Conclusion

Coplanarity does not make passive routing cheap. It changes the governing invariant. Once every target lies in a two-plane, the exact state count is the degree of an ordered projective rational interpolant. Fejér–Riesz lifting turns that algebraic object into a lossless network without spending an extra state, while a common-denominator argument proves that no smaller network can escape the plane. The result exposes a generic half-linear memory law in a regime where the optimized constant-detector hierarchy reports one, classifies the first nontrivial cross-ratio strata completely, and identifies the same integer as optimal integrated delay divided by 2π. More strongly, full-rank interpolation matrices impose an explicit nonzero chordal-error floor on every low-state lossless router, stable under calibrated perturbations. The strict four-node fixture turns the exact jump into a certified 21% one-state approximation barrier. The analysis also marks the next frontier precisely: higher-dimensional singular target varieties motivate detectors that move with frequency, rather than merely more constant minors.

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